

Skills Summary

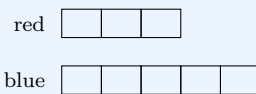
Tape Diagrams and Double Number Lines for Ratios

Use this reference while completing your worksheet or when studying.

Both diagrams do the same job: they keep two quantities scaling together. Find the value of *one* box, or the value at *one* unit, and the rest is multiplication.

1. Tape diagrams: find one box first

Rule: draw one box per part. Every box is worth the same amount, so value of one box = known amount $\div$ number of boxes it covers. Then multiply for the other quantity or for the total.



Example: Red to blue is 3 : 5. There are 12 red counters. How many blue?

Step 1. The known amount covers the red tape only, so start by splitting it over the 3 red boxes rather than over all the boxes.

Step 2. One box is $12 \div 3 = 4$ counters; blue is 5 boxes, so 5×4 .

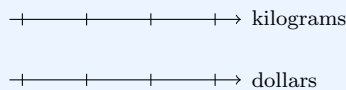
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Try it: Cats to dogs is 7 : 4. There are 21 cats. How many dogs?

check: 12

2. Double number lines: move both lines together

Rule: matching values sit above each other. Multiply or divide *both* lines by the same number to reach a new pair — never one line alone.



Example: 4 kilograms of flour cost 10 dollars. What do 6 kilograms cost?

Step 1. Six is not a whole number of steps from four, so go through one kilogram — the tick that makes every other value one multiplication.

Step 2. One kilogram is $10 \div 4 = 2.5$ dollars, so six kilograms is 6×2.5 .

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Try it: 6 tickets cost 18 dollars. What do 10 tickets cost?

check: 08

3. Reading the unit rate off the line

Rule: the unit rate is the number sitting opposite 1. Divide the known pair by the first quantity to get there. Once it is written on the line, every later question is “multiply by the unit rate”.

Example: 5 notebooks cost 45 dollars. Find the cost of one.

Step 1. The question asks for the value opposite 1 notebook, so divide both lines by 5.

Step 2. $45 \div 5 = 9$, and $5 \times 9 = 45$ checks it.

9 dollars per notebook

Try it: 8 pens cost 56 dollars. Find the cost of one pen.

check: $8 \times 7 = 56$

4. Comparing two ratios

Rule: two ratios can only be compared at the same place — the same number of one quantity. Use the unit rate, or write both as fractions with a common denominator. Bigger boxes do not mean a bigger ratio.

Example: Which is greater, $\frac{3}{4}$ or $\frac{5}{8}$?

Step 1. The denominators differ, so nothing can be compared until both parts are measured against the same whole.

Step 2. $\frac{3}{4} = \frac{6}{8}$, and 6 eighths is more than 5 eighths.

Try it: Compare $\frac{2}{3}$ and $\frac{5}{6}$.

check: $\frac{4}{6} > \frac{5}{6}$